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Category : A

(1)

$(x_1, y_1) = (120, 23)$ $\quad dy = y_2 - y_1 = 303$

$(x_2, y_2) = (423, 928)$ $\quad dx = x_2 - x_1 = 405$

$m = \frac{405}{303} = 1.32$

$|m| = 1.32 > 1$ So $x = x + \frac{1}{m}$. No. of x increm 1

Since, $|m| > 1$ increment y by 1

$\therefore |dy| + 1 = 405 + 1 = 406$ pixels

(2) Problem in DDA And solution

Floating-point problem. It's need extra hardware it's slower than integer math. If we use midpoint Algorithm then it enter solve.

round up error in line drifts when small errors add up. Solution use mid point algorithm

Direction issue where wrong pixels choosen

x_1, x_2) if we use Revised DDA then this problem can be solve.

(5) Aliasing is jagged edges caused by drawing smooth lines on a discrete pixel grid.

increasing resolution helps a little, but it's not a practical solution. Because it's too expensive. gives slow performance use more memory. and doesn't truly solve the problem that why we increase resolution to light aliasing.

Category B

Q a

(5,16) to (13,10)

$(x_1, y_1) = (5, 16)$

$(x_2, y_2) = (13, 10)$

$dy = 10 - 16 = -6$

$dx = 13 - 5 = 8$

$m = \frac{dy}{dx} = -0.75 \Rightarrow |M| < 1$

Since $|M| < 1$: increment x by 1 calculate $y_{k+1} = y_k + M$

Step	x	y	round(y)	Pixel
0	5	16.00	16	(5,16)
1	6	15.25	15	(6,15)
2	7	14.50	15	(7,15)
3	8	13.75	14	(8,14)
4	9	13.00	13	(9,13)
5	10	12.25	12	(10,12)
6	11	11.50	12	(11,12)
7	12	10.75	11	(12,11)
8	13	10.00	10	(13,10)

CamScanner

(b) (0,0) to (4,6)

$dx = 4 - 0 = 4$ $\quad m = \frac{dy}{dx} = 1.5 \Rightarrow |M| > 1$

$dy = 6 - 0 = 6$

Since $|M| > 1$ increment y by +1, calculate x_{k+1}

$= x_k + \frac{1}{m} = x_k + \frac{2}{3}$

Step	y	x	x Round	Pixel
0	0	0	0	(0,0)
1	1	0.667	1	(1,1)
2	2	1.334	1	(1,2)
3	2	2.001	2	(2,3)
4	4	2.668	3	(3,4)
5	5	3.335	3	(3,5)
6	6	4.002	4	(4,6)

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(10) $\frac{x}{7} - \frac{y}{12} = 5$

$\Rightarrow 12x - 7y = 420$

$y = 12$

$\Rightarrow 12x - 7(12) = 420$

$\Rightarrow 12x = 504$

$\Rightarrow x = 42$ $(42, 12)$

$y = 0$

$\Rightarrow 12x - 7(0) = 420$

$\Rightarrow 12x = 420$

$\Rightarrow x = 34$ $(35, 0)$

$dx = 35 - 42 = -7$

$dy = 0 - 12 = -12$

$|dx| < 0$, $dy < 0 \Rightarrow$ zone 5

Now, using 8-way symmetry

$(x, y) \rightarrow (-y, -x)$

$(42, 12) \rightarrow (-12, -42)$

$(35, 0) \rightarrow (0, -35)$

$dx = (0-(-12) = 12$

$dy = -35-(-42) = 7$

$d_{init} = 2dy - dx = 14 - 12 = 2$

$d_E = 2dy$
$= 2 \times 7$
$= 14$

$d_{NE} = 2dy - 2dx$
$= 2.(7-12)$
$= -10$

x	y	d	direction	dum	pixel (-y,-x)
-12	-42	2	NE	-8	42,12
-11	-41	-8	E	6	41, 11
-10	-41	6	NE	-4	41, 10
-9	-40	-4	E	10	40, 9
-8	-40	10	NE	0	40, 8
-7	-39	0	NE	-10	39, 7